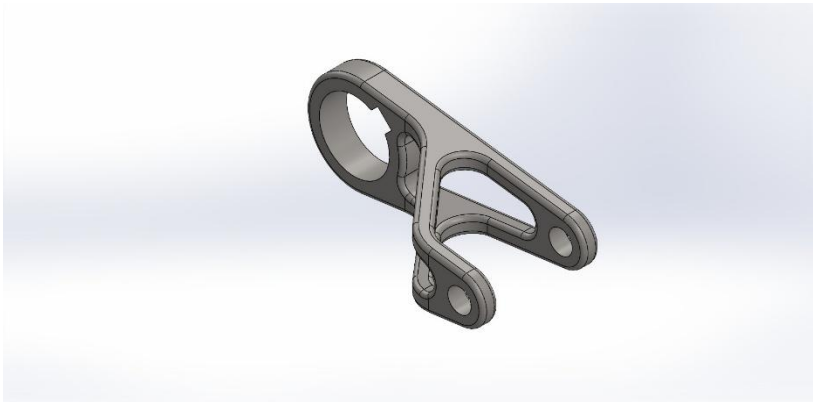




Simulation of Model_Mania_2010

Date: Hidden
Designer: Solidworks
Study name: SimulationXpress Study
Analysis type: Static

Table of Contents

Description	1
Assumptions.....	2
Model Information	2
Material Properties	3
Loads and Fixtures	3
Mesh information.....	4
Study Results	5

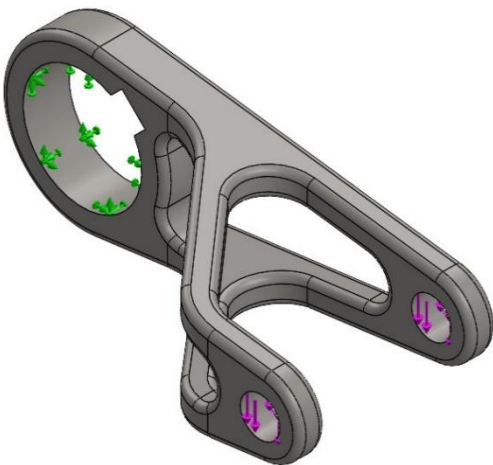
Description

Fixture and Forces were defined as asked for in Phase 2 Instructions.



Assumptions

Model Information



Model name: Model_Mania_2010
Current Configuration: Phase 2

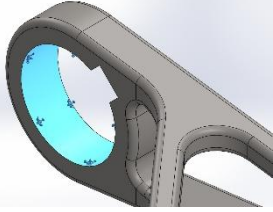
Solid Bodies			
Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Fillet2 	Solid Body	Mass:0.328447 kg Volume:4.15759e-05 m^3 Density:7,899.95 kg/m^3 Weight:3.21879 N	Hidden

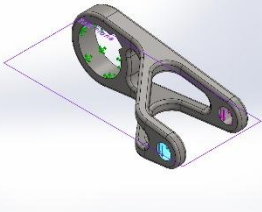
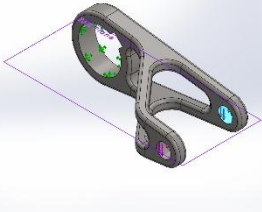


Material Properties

Model Reference	Properties	Components
	Name: AISI 1020 Model type: Linear Elastic Isotropic Default failure criterion: Max von Mises Stress Yield strength: 3.51571e+08 N/m ² Tensile strength: 4.20507e+08 N/m ²	SolidBody 1(Fillet2)(Model_Mania_2010)

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
Fixed-2		Entities: 1 face(s) Type: Fixed Geometry

Load name	Load Image	Load Details
Force-3		Entities: 1 face(s), 1 plane(s) Reference: Top Plane Type: Apply force Values: ---, ---, -, 500 N
Force-4		Entities: 1 face(s), 1 plane(s) Reference: Top Plane Type: Apply force Values: ---, ---, -, 500 N



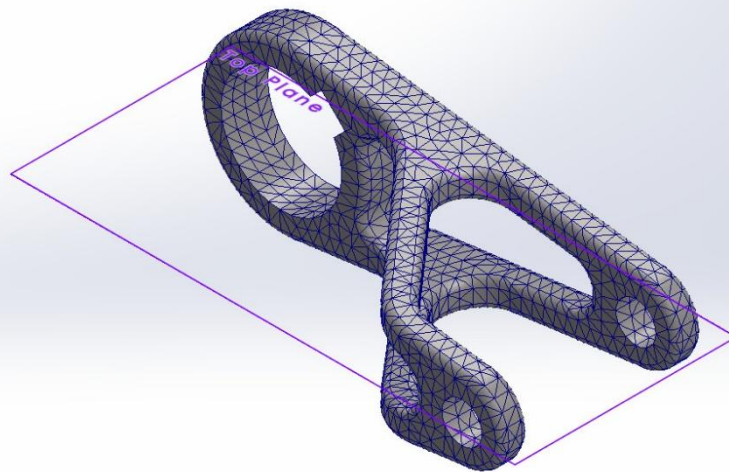
Mesh information

Mesh type	Solid Mesh
Mesher Used:	Standard mesh
Automatic Transition:	Off
Include Mesh Auto Loops:	Off
Jacobian points for High quality mesh	16 Points
Element Size	3.46543 mm
Tolerance	0.173271 mm
Mesh Quality	High

Mesh information - Details

Total Nodes	15061
Total Elements	8670
Maximum Aspect Ratio	6.658
% of elements with Aspect Ratio < 3	98.8
Percentage of elements with Aspect Ratio > 10	0
Percentage of distorted elements	0
Time to complete mesh(hh:mm:ss):	00:00:06
Computer name:	Hidden

Model name: Model_Mania_2010
Study name: SimulationXpress Study(-Phase 2-)
Mesh type: Solid Mesh



SOLIDWORKS Educational Product. For Instructional Use Only.



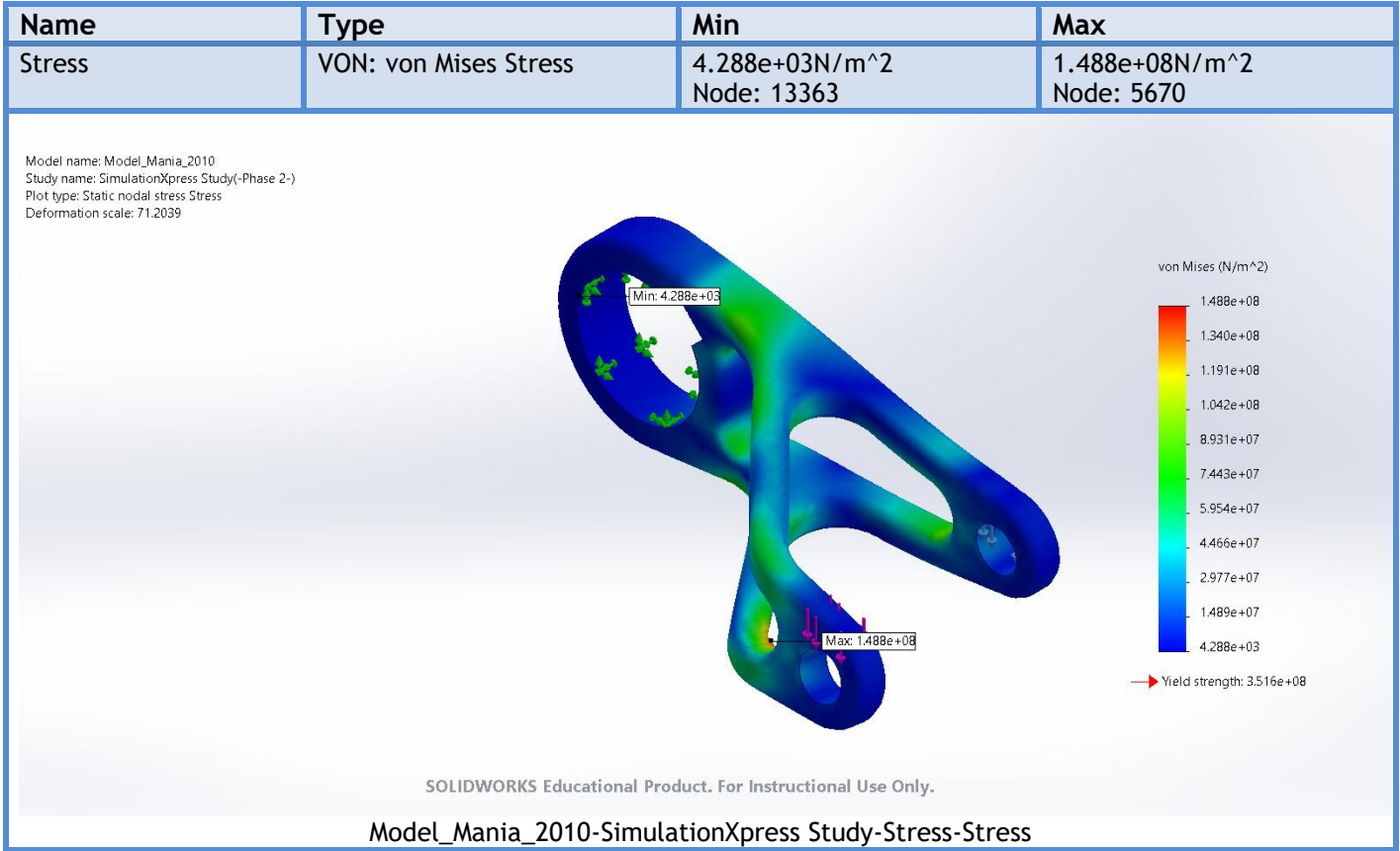
SOLIDWORKS

Analyzed with SOLIDWORKS Simulation

Simulation of Model_Mania_2010

4

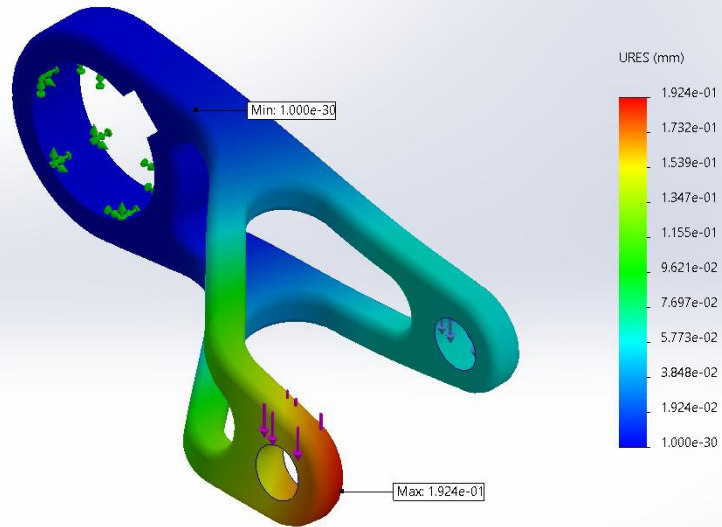
Study Results



Name	Type	Min	Max
Displacement	URES: Resultant Displacement	0.000e+00mm Node: 286	1.924e-01mm Node: 55



Model name: Model_Mania_2010
 Study name: SimulationXpress Study(-Phase 2-)
 Plot type: Static displacement Displacement
 Deformation scale: 71.2039

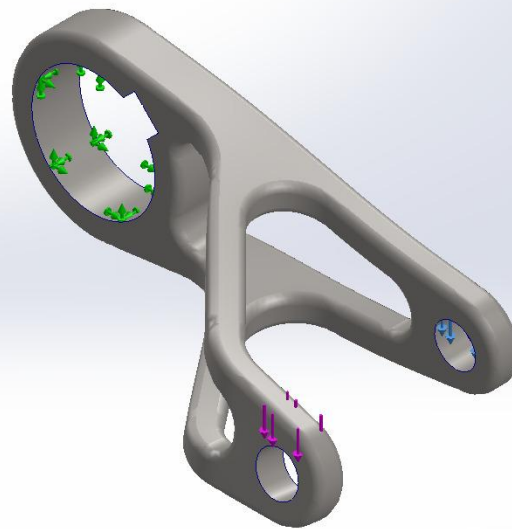


SOLIDWORKS Educational Product. For Instructional Use Only.

Model_Mania_2010-SimulationXpress Study-Displacement-Displacement

Name	Type
Deformation	Deformed shape

Model name: Model_Mania_2010
 Study name: SimulationXpress Study(-Phase 2-)
 Plot type: Deformed shape Deformation
 Deformation scale: 71.2039



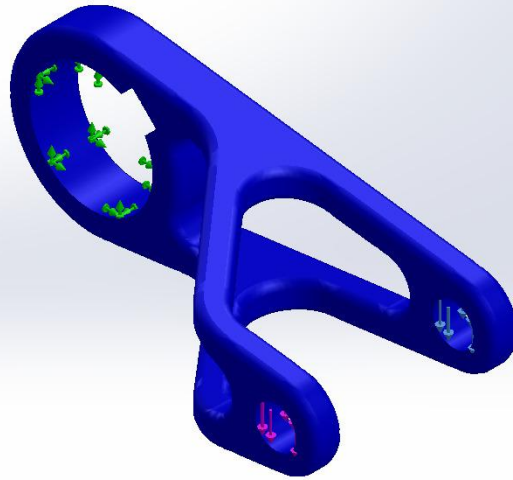
SOLIDWORKS Educational Product. For Instructional Use Only.

Model_Mania_2010-SimulationXpress Study-Displacement-Deformation

Name	Type	Min	Max
Factor of Safety	Max von Mises Stress	2.362e+00	8.199e+04



Model name: Model_Mania_2010
Study name: SimulationXpress Study(-Phase 2-)
Plot type: Factor of Safety Factor of Safety
Criterion: Max von Mises Stress
Red < FOS = 1 < Blue



SOLIDWORKS Educational Product. For Instructional Use Only.

Model_Mania_2010-SimulationXpress Study-Factor of Safety-Factor of Safety

